RAGAVI RAJA

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Tiruchengode, Namakkal(DT)

Professional Summary

As a fresher in the field of Biomedical Engineering, I am open to learning in a demanding environment. Having skills in PLC programming, SCADA, CX-Programmer, industrial automation, python programming, and embedded system, I am eager to make use of my skills and have industry exposure in the field of automation and controls.

Technical Skills

- **Programming Language:** Python , Embedded C
- **Automation:** Siemens PLC, Allen-Bradley PLC, Omron PLC, SCADA Ladder logic programming (CX Programmer, CX Simulator).
- **Tools:** Power BI, MATLAB, Numpy, Pandas, Proteus, Omron software ,Blender , LabVIEW.
- **Core skills :**Medical Imaging and signal processing communications, ECG Analysis, Cardiac Conduction, Vital Signs Monitoring.
- **Others:** Office Automation, MS Excel(Advance level)

Internships

CADD Centre | Software Trainer — Present-2026
Trained students in python, PLC, SCADA, software applications.
Conducted practical sessions, workshops, and project-based learning activities.

Healthwatch Telediagnosis pvt.ltd – Salem — Jan-2026
Monitored and analysed real-time ECG waveforms and heart rate data for cardiac assessment.
Assisted in identifying ECG patterns, abnormalities, and arrhythmia indications under supervision.

Vinayaka Mission Multispeciality Hospital | Biomedical Intern — July- 2025
Gained hands-on exposure to biomedical equipment used in hospital environments.
Assisted in equipment monitoring, quality checks, and technical documentation.

ITECH Multi speciality Hospital Salem | Biomedical Intern — October-2024
Assisted in the maintenance and troubleshooting of biomedical equipment.
Observed the operation of ECG, patient monitors, infusion pumps, and other medical devices.

Education

BE.BIOMEDICAL ENGINEERING
Sri Shanmugha College of
Engineering and Technology
Pullipalayam , Salem (DT)
Year of completion : 2026
CGPA : 8.2

Certifications

- Office Automations – Livewire CADD centre
- Course on Python – Livewire CADD centre
- Internet of Things(IOT) – NPTEL
- Healthcare Analytics – Coursa
- Bioinformatics – Coursa

Projects

Title : Smart Rehabilitation System Using EMG Signals

Developed a Smart Rehabilitation System using EMG signals to monitor muscle activity during rehabilitation exercises. Acquired and processed EMG signals using surface electrodes and signal conditioning circuits. Implemented real-time muscle contraction and fatigue detection through signal analysis techniques. Designed a user-friendly interface to visualize EMG data and track patient recovery progress. Integrated hardware and software components to provide a portable, cost-effective rehabilitation monitoring solution.

Mini Project : TENS Machine Stimulation

Developed a Transcutaneous Electrical Nerve Stimulation (TENS) machine that uses electrical pulses for pain relief. The machine was designed with the intention of designing and building pulse generation and control circuits. I performed testing and validation of the output parameters such as frequency, pulse width, and intensity. Through this project, my technical skills were sharpened in areas such as biomedical instrumentation, electronic circuit design, and signal generation.